

Han Joon Byun

Researcher in Machine Learning, Optimization, and Finance

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RESEARCH PROFILE

Researcher working at the intersection of machine learning, mathematical optimization, and finance, with a focus on tabular data and time-series modeling. Current research combines evolutionary computation and Transformer architectures to improve predictive performance, computational efficiency, and interpretability. Brings rigorous mathematical training and applied experience spanning financial factors, credit-risk prediction, and interpretable AI.

Research areas: Tabular machine learning · Time series · Evolutionary computation · Transformer models · Financial AI · Interpretable machine learning

EDUCATION

Ph.D. in Computer Science and Engineering <i>Seoul National University · Optimization Lab</i>	Expected 2027
M.S. in Computer Science and Engineering <i>Seoul National University · Optimization Lab</i>	2024
B.S. in Mathematics <i>New York University</i> <i>Advanced study included Probability Theory, Statistics, and Stochastic Calculus.</i>	2018

RESEARCH & APPLIED EXPERIENCE

Agency for Defense Development · Joint Research Project • Contributed to research on interpretable modeling, emphasizing methods that make model behavior easier to understand.	2026
PFCT · Joint Research Project • Developed a default-prediction model using tabular Transformer architectures for structured financial data.	2025
Think Pool · Joint Research Project • Developed AI-driven financial factors, applying machine-learning methods to quantitative finance.	2022–2024
Republic of Korea Air Force · Interpretation Officer • Served as an interpretation officer and established a long-term career direction at the intersection of computer science and finance.	2019–2022

TEACHING EXPERIENCE

Head Teaching Assistant · Data Structures <i>Seoul National University</i>	Spring 2023 · Fall 2024 · Fall 2025
Head Teaching Assistant · Algorithms <i>Seoul National University</i>	Fall 2023 · Spring 2025 · Spring 2026

Prepared course content, coordinated teaching support, answered student questions, and graded assignments and examinations. Maintained an approachable, hands-on presence to help students build confidence in core problem-solving concepts.

RESEARCH TRAJECTORY

Neural Network Foundations Studied CNN and ANN pruning and transfer learning to understand model learning and efficiency.	Combinatorial Optimization Helped develop a bespoke Transformer that frames the Traveling Salesperson Problem through language modeling.	Financial & Tabular AI Applies optimization and neural methods to improve performance, efficiency, and interpretability.
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PUBLICATIONS

1.

RTSC: Return-Aware Token-Space Contrastive Transformer for Cross-Sectional Stock Return Prediction
Hanbin Koo^{*}, **Han Joon Byun**^{*}, and Byung-ro Moon
^{*} *Equal contribution.*
Submitted to ICAIF 2026 · **Under review**

2.

Certificates, Declarations, and Abstention After Offline Policy Selection: Two Synthetic Operating-Characteristic Calibrations
Kihun Rhee and **Han Joon Byun**
Submitted to Transactions on Machine Learning Research (TMLR) · **Under review**

3.

Label-Noise Decay Confounds Loss-Based Drift Attribution: A False-Positive Channel and a Noise-Aware Repair
Huh Joon and **Han Joon Byun**
Submitted to Transactions on Machine Learning Research (TMLR) · **Under review**

4.

Multi-Objective Evolution of Patch Tokens for Tabular Transformers
Han Joon Byun, Joon Huh, Jieun Yook, and Byung-ro Moon
Proceedings of the Genetic and Evolutionary Computation Conference (GECCO '26), 2026
doi.org/10.1145/3795095.3805076

5.

Evolutionary Feature Construction for Tabular Transformers via Gradient-Enhanced Attention Rollout
Han Joon Byun, Jieun Yook, Hanbin Koo, and Byung-ro Moon
Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '26 Companion), 2026
doi.org/10.1145/3795101.3805282

6.

CycleFormer: TSP Solver Based on Language Modeling
Jieun Yook, Junpyo Seo, Joon Huh, **Han Joon Byun**, and Byung-ro Moon
arXiv preprint, 2024
arxiv.org/abs/2405.20042

7.

Evolutionary Pruning of Deep Convolutional Networks by a Memetic GA with Sped-Up Local Optimization and GLCM Energy Z-Score
Ha Hyun Cho, **Han Joon Byun**, Min Kee Kim, Joon Huh, and Byung-Ro Moon
Proceedings of the Genetic and Evolutionary Computation Conference Companion (GECCO '23 Companion), 2023
doi.org/10.1145/3583133.3590604